

# **Visual Causal-chain Bookmarking: Retrieval-Weighted Credit Assignment for Reinforcement Learning from Delayed Reward**

A dissertation submitted to The University of Manchester for the degree of  
**Master of Science in Robotics**  
in the Faculty of Science and Engineering

**Year of submission**

2026

**Student ID**

14354077

School of Engineering

# Contents

|                                                                                         |            |
|-----------------------------------------------------------------------------------------|------------|
| <b>Contents</b>                                                                         | <b>I</b>   |
| <b>Abstract</b>                                                                         | <b>IV</b>  |
| <b>Declaration of originality</b>                                                       | <b>V</b>   |
| <b>Copyright statement</b>                                                              | <b>VI</b>  |
| <b>Acknowledgments</b>                                                                  | <b>VII</b> |
| <b>1 Introduction</b>                                                                   | <b>1</b>   |
| <b>2 Preliminaries</b>                                                                  | <b>1</b>   |
| 2.1 Markov decision processes and policy-gradient reinforcement learning                | 1          |
| 2.2 RUDDER: LSTM-based return redistribution                                            | 2          |
| 2.3 LOOP: leave-one-out advantage estimation with PPO-clipped updates                   | 3          |
| <b>3 Related work</b>                                                                   | <b>3</b>   |
| 3.1 Temporal and counterfactual credit assignment beyond RUDDER                         | 3          |
| 3.2 Group-relative and trajectory-level RL fine-tuning beyond LOOP                      | 4          |
| 3.3 Episodic memory and retrieval in long-horizon LLM agents                            | 4          |
| 3.4 Surveys establishing the scope of the credit-assignment problem                     | 5          |
| 3.5 VCBM’s distinguishing position                                                      | 5          |
| <b>4 Technical contribution</b>                                                         | <b>5</b>   |
| 4.1 VCBM for tabular MDPs: causal discovery and Welford estimation                      | 6          |
| 4.1.1 Motivation                                                                        | 6          |
| 4.1.2 Causal-role discovery                                                             | 6          |
| 4.1.3 Bookmark discovery and value estimation                                           | 7          |
| 4.1.4 Action selection                                                                  | 7          |
| 4.2 VCBM for language-based MDPs: retrieval-weighted credit blending                    | 7          |
| 4.2.1 Motivation                                                                        | 7          |
| 4.2.2 Bookmark detection and episodic memory                                            | 8          |
| 4.2.3 Retrieval and temperature-scaled weighting                                        | 8          |
| 4.2.4 Blending with the trajectory-level LOO advantage                                  | 8          |
| 4.3 Where the two instantiations agree and where they necessarily diverge               | 8          |
| 4.4 Six structural capabilities VCBM has that a uniform trajectory-level baseline lacks | 9          |
| <b>5 Experiments</b>                                                                    | <b>10</b>  |

|          |                                                                                  |           |
|----------|----------------------------------------------------------------------------------|-----------|
| 5.1      | Controlled evaluation: RUDDER vs. VCBM in a delayed-reward tabular MDP . . . . . | 10        |
| 5.1.1    | Environment . . . . .                                                            | 10        |
| 5.1.2    | Method and metrics . . . . .                                                     | 10        |
| 5.1.3    | Discovered causal structure . . . . .                                            | 11        |
| 5.1.4    | Interpretation . . . . .                                                         | 11        |
| 5.2      | Large-scale evaluation: LOOP vs. VCBM-blended LOOP on AppWorld . . . . .         | 12        |
| 5.2.1    | Setup . . . . .                                                                  | 12        |
| 5.2.2    | Task performance . . . . .                                                       | 12        |
| 5.2.3    | Off-policy drift and optimisation stability . . . . .                            | 13        |
| 5.2.4    | Efficiency . . . . .                                                             | 13        |
| 5.2.5    | Parity and open questions . . . . .                                              | 14        |
| 5.2.6    | Interpretation . . . . .                                                         | 14        |
| <b>6</b> | <b>Conclusions . . . . .</b>                                                     | <b>14</b> |
|          | <b>References . . . . .</b>                                                      | <b>16</b> |
|          | <b>Appendices . . . . .</b>                                                      | <b>22</b> |
| <b>A</b> | <b>Project Outline . . . . .</b>                                                 | <b>22</b> |
| <b>B</b> | <b>Risk Assessment . . . . .</b>                                                 | <b>24</b> |

**Word count: 4833**

## Abstract

Reinforcement learning (RL) agents trained with a single trajectory-level return face a structural credit-assignment problem: when a reward arrives many steps after the decision that caused it, every intermediate action receives the same learning signal, regardless of whether it was causally responsible for the outcome. This dissertation studies *Visual Causal-chain Bookmarking* (VCBM), a family of methods that concentrate credit on the specific decision points that determine an episode’s return rather than spreading it uniformly across the trajectory, and develops two domain-specific instantiations of this idea.

The first instantiation targets a fully observed, low-dimensional Markov decision process (a delayed-reward watch-repair task), where VCBM statistically discovers which state components are causally controlled by the agent’s actions and estimates per-decision-point action values directly; because the environment’s causal structure is verifiable, this instantiation is evaluated against known ground truth as well as against RUDDER [1], a return-redistribution baseline built on an LSTM contribution network. VCBM converges to a 95.07% correct-decision rate in a mean of  $1,532.3 \pm 1,428.0$  episodes across 20 random seeds (range 1,196–7,757), against RUDDER’s slower and more variable convergence (mean  $3,327.1 \pm 2,010.9$  episodes, median 2,829, range 1,603–10,936, computed from the 20 per-seed training logs); this difference is statistically significant (unpaired Mann–Whitney  $U = 19$ ,  $p < 0.001$ , rank-biserial  $r = 0.91$ ), and VCBM’s discovered causal structure exactly matches the environment’s true generative structure in all 20 seeds.

The second instantiation targets a partially observed, language-based setting: an LLM agent trained with Leave-One-Out Proximal Policy Optimisation (LOOP) [2] inside the AppWorld benchmark [3], where causal discovery is intractable and a structural heuristic (state-mutating tool calls) substitutes for it, with credit distributed via cosine-similarity retrieval over an episodic memory of bookmarked turns and blended with LOOP’s trajectory-level advantage. Trained for 200 iterations on identical Qwen2.5-7B-Instruct/LoRA/FSDP2 configurations, VCBM-blended LOOP shows an exact improvement over plain LOOP, taken directly from the W&B summary export for both runs, on task success rate (0.354 vs. 0.458 overall, a +29.4% change concentrated almost entirely in medium-difficulty tasks: 0.188 vs. 0.500, +166.7%), on the final-step maximum importance weight (−22.2%), and on output-token efficiency (−18.8% mean output tokens, +34.6% return per 1,000 tokens); the final-step KL estimate is a marginal exception, moving 4.1% in LOOP’s favour. This AppWorld comparison remains a single-seed, final-step result for each method; unlike the tabular result, it has not been tested for statistical significance across multiple seeds (Section 5.2).

Taken together, the watch-repair result supports the dissertation’s central claim, that explicitly identifying and weighting the causally relevant steps of a trajectory is a more sample-efficient credit-

assignment strategy than either an undifferentiated trajectory-level baseline or a learned return-redistribution network, and the AppWorld result is consistent with the same claim on an exact, single-seed comparison, though it has not been tested for statistical significance across multiple seeds in the way the tabular result has.

## **Declaration of originality**

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

## Copyright statement

- i The author of this thesis (including any appendices and/or schedules to this thesis) owns certain copyright or related rights in it (the “Copyright”) and s/he has given The University of Manchester certain rights to use such Copyright, including for administrative purposes.
- ii Copies of this thesis, either in full or in extracts and whether in hard or electronic copy, may be made *only* in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has from time to time. This page must form part of any such copies made.
- iii The ownership of certain Copyright, patents, designs, trademarks and other intellectual property (the “Intellectual Property”) and any reproductions of copyright works in the thesis, for example graphs and tables (“Reproductions”), which may be described in this thesis, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.
- iv Further information on the conditions under which disclosure, publication and commercialisation of this thesis, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy (see <http://documents.manchester.ac.uk/DocuInfo.aspx?DocID=24420>), in any relevant Thesis restriction declarations deposited in the University Library, The University Library’s regulations (see <http://www.library.manchester.ac.uk/about/regulations/>) and in The University’s policy on Presentation of Theses.

## **Acknowledgments**

I would like to thank my parents, Geetha Selvakumar and Selvakumar Palaniyandi, for their constant support throughout this degree. Thanks also to my younger brother, Saran Selvakumar, who has never once let me take myself too seriously, and to my friends who pushed me to keep going when this project got difficult.

I am grateful to my supervisor, Dr. Pawel, for their guidance throughout this project. A special thanks to Swetha Parthasarathy, a good friend, whose encouragement pushed me to go further than I thought I could.

# 1 Introduction

Reinforcement learning (RL) agents that learn from a trajectory-level reward face a structural credit-assignment problem whenever that reward is *delayed* and *sparse*: it arrives only at the end of an episode, after many intermediate actions, most of which did not causally influence the outcome. This recurs from discrete-action control [4], through robotic control [5], [6], to RLHF post-training of LLMs [7], [8], [9]: an LLM agent solving a task in AppWorld [3] executes a long tool-call sequence of which only a handful change task-relevant state, yet a policy-gradient update that assigns one scalar advantage to every action averages the delayed signal across all of them, slowing convergence and risking spurious action-outcome correlations [10], [11].

Two established families of methods address this without questioning the underlying strategy. Trajectory-level baselines, such as the leave-one-out estimator used in LOOP [2], reduce return variance by subtracting a baseline from other rollouts of the same scenario, but every token still receives an identical advantage. Return-redistribution methods such as RUDDER [1] instead train an LSTM to predict the eventual return from every trajectory prefix and redistribute reward as the difference between consecutive predictions: denser in principle, but only as reliable as the LSTM’s own convergence.

An underexploited alternative is to make credit assignment *structural*: identify which timesteps are causally responsible and concentrate the signal there. This is the idea behind *Visual Causal-chain Bookmarking* (VCBM), instantiated two ways. In a fully observed, low-dimensional MDP, a hypothesis test determines which state components an action causally controls, so VCBM performs explicit *causal discovery* and accumulates return statistics per bookmarked decision via a Welford estimator. In a partially observed, natural-language setting such as AppWorld, this discovery is intractable, so the second instantiation substitutes a structural heuristic (state-mutating tool calls) and retrieval over an episodic memory, blended with the trajectory-level LOO advantage.

This dissertation designs, implements, and evaluates both instantiations against their natural baselines. Section 2 formalises the baselines and Section 3 situates VCBM in the literature. Section 4 then gives the full formulation, before Section 5 reports the tabular and AppWorld comparisons and Section 6 closes with what was achieved and where it falls short.

## 2 Preliminaries

### 2.1 Markov decision processes and policy-gradient reinforcement learning

An episodic MDP [12] is a tuple  $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$  with state space  $\mathcal{S}$ , action space  $\mathcal{A}$ , transition kernel  $P(s_{t+1} \mid s_t, a_t)$ , reward function  $r_t = r(s_t, a_t)$ , discount factor  $\gamma \in (0, 1]$ , and horizon  $T$ . A stochastic

policy  $\pi_\theta(a_t \mid s_t)$ , parameterised by  $\theta$ , generates a trajectory  $\tau = (s_0, a_0, r_0, \dots, s_T)$  with realised (undiscounted, as used throughout this dissertation’s environments) return

$$G(\tau) = \sum_{t=0}^T r_t. \quad (1)$$

Policy-gradient methods [13] maximise  $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[G(\tau)]$  by ascending  $\nabla_\theta J(\theta) = \mathbb{E}_\tau[\sum_t \nabla_\theta \log \pi_\theta(a_t \mid s_t) A_t]$ , where  $A_t$  is an *advantage* that reduces gradient variance relative to using  $G(\tau)$  directly, following the baseline-subtraction principle that leaves the optimal policy invariant [14] and can be estimated recursively [15]. That a late reward gives a weak, high-variance signal for early decisions is a long-standing obstacle for trial-and-error learning [16]. RUDDER and LOOP, the two baselines studied here, both address this variance problem but with different mechanisms; this dissertation’s central hypothesis, tested via VCBM, is that a third mechanism, explicit identification of the causally relevant timesteps, offers a more sample-efficient path to a low-variance  $A_t$  than either.

## 2.2 RUDDER: LSTM-based return redistribution

RUDDER [1] addresses delayed, sparse reward by training an auxiliary sequence model to predict the eventual episode return from every trajectory prefix, then using the *change* in that prediction between consecutive timesteps as a redistributed, per-step reward. Let  $\Delta s_t = s_t - s_{t-1}$  denote the state-difference at step  $t$ . An LSTM [17] with disabled forget and output gates produces a return prediction at every prefix,

$$\hat{G}(s_{0:t}, a_{0:t}) = \text{Linear}(\text{LSTM}(\Delta s_{0:t}, a_{0:t}))_t, \quad (2)$$

trained against the true (scaled) episode return  $G_0 = G(\tau)/R_{\text{scale}}$  with a combination of a terminal-step loss and an auxiliary loss applied at every prefix,

$$\mathcal{L}_{\text{RUDDER}} = \underbrace{(\hat{G}(s_{0:T}) - G_0)^2}_{\mathcal{L}_{\text{main}}} + \lambda_{\text{aux}} \underbrace{\frac{1}{T} \sum_{t=0}^T (\hat{G}(s_{0:t}) - G_0)^2}_{\mathcal{L}_{\text{aux}}}. \quad (3)$$

The redistributed reward at each timestep is then the scaled first difference of consecutive predictions,

$$\tilde{r}_t = R_{\text{scale}} \cdot (\hat{G}(s_{0:t}) - \hat{G}(s_{0:t-1})), \quad \hat{G}(s_{0:-1}) \equiv 0, \quad (4)$$

which sums telescopically to the true episode return,  $\sum_t \tilde{r}_t = R_{\text{scale}} \cdot \hat{G}(s_{0:T}) \approx G(\tau)$ , while ideally concentrating reward at timesteps the LSTM has learned are predictive of the outcome. RUDDER’s redistribution is thus a *learned, statistical* solution: it approximates causal structure via function approximation rather than identifying it directly, and is used here as the representative baseline against which VCBM’s structural approach is compared in a tabular setting (Section 5).

## 2.3 LOOP: leave-one-out advantage estimation with PPO-clipped updates

LOOP [2] addresses the same problem in a different regime: training an LLM policy via multi-turn tool-use rollouts, where a value-function critic as used in actor-critic [18] or trust-region [19] methods is memory-expensive alongside a large language model. Given  $K$  rollouts  $\tau^{(1)}, \dots, \tau^{(K)}$  from the same scenario, each with return  $G^{(k)} = G(\tau^{(k)})$ , the leave-one-out baseline for rollout  $k$  [20] is the mean return of the *other*  $K - 1$  rollouts,

$$b^{(k)} = \frac{1}{K-1} \sum_{j \neq k} G^{(j)}, \quad A^{(k)} = G^{(k)} - b^{(k)}, \quad (5)$$

which is applied uniformly to every token generated within rollout  $k$ , i.e.  $A_t = A^{(k)}$  for all  $t$  in that rollout: the trajectory-level credit-assignment limitation that VCBM’s blend mechanism (Section 4) directly targets.

This uniform advantage is consumed by a PPO-clipped policy-gradient objective, needed because rollouts are generated by a separate, possibly numerically divergent inference engine. The update requires an importance weight correcting for the mismatch between the sampling log-probability  $\pi_{\theta_{\text{old}}}$  and the current training log-probability  $\pi_{\theta}$ :  $w_t = \exp(\log \pi_{\theta}(a_t) - \log \pi_{\theta_{\text{old}}}(a_t))$ . The per-token objective applies a PPO-style clip [21] to bound any single update,

$$\mathcal{L}_{\text{PPO}}(t) = -\min(w_t A_t, \text{clip}(w_t, 1 - \epsilon, 1 + \epsilon) A_t), \quad (6)$$

with clip range  $\epsilon$  (default 0.1 here). Equation (6) is the loss into which VCBM’s retrieval-weighted, per-token advantage (Section 4.2) is substituted in place of the uniform  $A_t = A^{(k)}$ : VCBM modifies *which* advantage each token receives, with no change to the surrounding PPO-clipped machinery.

## 3 Related work

This section situates VCBM against the wider credit-assignment literature that motivates its design but is not directly benchmarked in Section 5, in four groups: temporal/counterfactual methods beyond RUDDER; group-relative/trajectory-level RL fine-tuning beyond LOOP; episodic memory and retrieval for LLM agents; and surveys of the problem’s scope.

### 3.1 Temporal and counterfactual credit assignment beyond RUDDER

RUDDER (Section 2.2) belongs to a broader family that redistributes reward by learning an auxiliary predictive model rather than testing for causal structure directly. Hindsight Credit Assignment (HCA) [22] learns a hindsight distribution over which action, conditioned on an observed outcome, was responsible, and rewrites the value function accordingly, a *learned, statistical* redistribution like

RUDDER’s LSTM. Counterfactual Credit Assignment (CCA) [23] conditions a value function on future events via a learned future-conditional baseline; COCOA [24] builds on this but measures contribution against a learned representation of the rewarding *event* rather than the rewarding state, reducing HCA’s spurious-credit failure mode. CCA and COCOA are conceptually closer to VCBM’s causal framing, since both pose a counterfactual question, but both require a learned forward/counterfactual model to answer it, whereas VCBM’s tabular test (Equation (7)) needs only online frequency counts, trading generality for a test that is exact and directly verifiable (Section 5.1). Randomized Return Decomposition [25] and Counterfactual Shapley Credit Assignment [26] (a Monte-Carlo subsequence-sampling objective and a Shapley-value attribution grounded in a structural causal model, respectively) likewise redistribute credit continuously over every timestep; VCBM’s departure is that a timestep either is or is not a bookmark, with no residual credit in between.

### 3.2 Group-relative and trajectory-level RL fine-tuning beyond LOOP

LOOP’s leave-one-out advantage (Section 2.3) is one instance of a broader class computing a single scalar advantage per trajectory, applied uniformly to every token. Turn-level reward design for multi-turn reasoning [27] extends GRPO/PPO with a dense, per-turn reward: the same coarse-granularity limitation as VCBM addresses, but by *redesigning the reward* at every turn rather than retrieving and weighting a discrete subset afterward. G2PO [28] is closer, being evaluated on AppWorld directly: it builds a global state-transition graph across rollouts and standardises temporal-difference errors on it to identify progress-driving transitions, exploiting state-transition structure much as VCBM’s bookmark rule does. The distinction is mechanism: G2PO reads credit off a learned, cross-rollout graph, whereas VCBM’s AppWorld instantiation uses a simple structural heuristic (state-mutating tool call) plus flat embedding-similarity retrieval, trading G2PO’s richer structure for a mechanism cheap enough to run inline with no graph construction. SPA-RL [29] densifies rewards via a learned stepwise-progress estimator, non-uniform like VCBM, but attributing *progress* at every step rather than selecting a discrete causal subset; VCBM’s bookmark mechanism is sparse where SPA-RL’s is dense.

### 3.3 Episodic memory and retrieval in long-horizon LLM agents

VCBM’s retrieval mechanism (Equation (11)) draws on using external memory to inform credit assignment. Memory-R2 [30] is the closest comparison: it addresses fair credit for memory-augmented LLM agents whose rollouts diverge because they write, update, or delete different memories, via LoGo-GRPO, which combines a global trajectory-level objective with local rerollouts from shared intermediate memory states, architecturally close to VCBM’s blend of a bookmark memory with a trajectory-level advantage (Equation (12)). The distinction: LoGo-GRPO’s fairness correction oper-

ates *within* the group-relative advantage computation, changing which rollouts are compared, whereas VCBM keeps memory-weighting and the LOOP objective as two separable stages joined only by  $\beta$ . Memory-R2’s setting is also explicitly *multi-session*, while VCBM’s AppWorld instantiation retrieves within/across single-episode rollouts of one training run, a related but not identical failure mode; this dissertation does not claim VCBM is a special case of LoGo-GRPO, only that both belong to the same family of memory-conditioned credit assignment.

### 3.4 Surveys establishing the scope of the credit-assignment problem

Four surveys help frame where VCBM sits in the literature. Pignatelli et al. [31] survey temporal credit assignment in deep RL broadly, with a formalism this dissertation’s tabular experiment is consistent with in spirit. Two further surveys narrow the scope to LLM agents: one [32] identifies credit assignment as a central bottleneck for scaling LLM agents to long-horizon settings, and a companion [33] organises 47 methods by granularity (token- to multi-agent-level) and methodology, noting that outcome-only signals grow uninformative as horizon lengthens. A fourth [34] reviews the full long-horizon agent lifecycle, situating credit assignment as one stage among several. Together they indicate the field has largely converged on *learned* solutions to the granularity problem that VCBM instead addresses, in its tabular instantiation, with an explicit statistical test.

### 3.5 VCBM’s distinguishing position

Across all three groups above, most surveyed methods assign credit through a *learned or continuously-valued* mechanism. VCBM’s two instantiations are deliberately harder-edged: in the tabular case a bookmark either is or is not causally controlled (fixed-threshold hypothesis test, no smoothing); in the AppWorld case what counts as a bookmark is a structural rule rather than a learned classifier, though *weighting* is soft (Equation (11)), closest to memory-augmented methods like Memory-R2. CCA and COCOA share VCBM’s causal framing without its hardness, substituting a learned counterfactual model. This dissertation’s contribution is thus best read not as introducing causal or memory-based credit assignment as a category, but as testing how far a hard, discovery-or-heuristic-driven bookmark selection can go before the softer, learned mechanisms that dominate the literature become necessary.

## 4 Technical contribution

Both VCBM instantiations share a three-stage template (*detect* decision points that plausibly carry causal weight, *store* them in structured memory, and *weight* the learning signal accordingly), but the mechanism at each stage differs between a fully observed tabular MDP, where causal discovery is

statistically tractable, and a partially observed, language-based MDP, where it is not. This section presents both in full and states explicitly where they diverge.

## 4.1 VCBM for tabular MDPs: causal discovery and Welford estimation

### 4.1.1 Motivation

In a fully observed MDP with a small, fixed-dimensional state space (the kind of structured representation that contrastive [35] and recurrent [36] world models otherwise learn from raw observations), it is possible to test directly, from interaction data alone, which state components an action causally influences. Credit for the final return can then be assigned exactly to the decision points where a controlled change originates, rather than diffused across an entire trajectory of filler steps, as in the watch-repair environment of Section 5, where a single decision at  $t = 0$  determines the return realised 50 steps later.

### 4.1.2 Causal-role discovery

Let the state decompose into  $n$  scalar components  $s = (s^{(1)}, \dots, s^{(n)})$ . For each component  $i$  and action  $a \in \{0, 1\}$ , a running count  $c_{i,a}$  tracks how often  $i$  changed value on a step with action  $a$ , out of  $n_a$  such steps. After a warm-up period, each component is classified by its empirical change rate  $\rho_{i,a} = c_{i,a}/n_a$ :

$$\text{role}(i) = \begin{cases} \text{CLOCK} & \min(\rho_{i,0}, \rho_{i,1}) > \eta_{\text{clock}} \\ \text{STATIC} & c_{i,0} + c_{i,1} = 0 \\ \text{CONTROLLED} & z_i > z_{\text{thresh}} \\ \text{NOISE} & \text{otherwise,} \end{cases} \quad (7)$$

where  $z_i$  is the two-proportion test statistic comparing the change rate under each action,

$$z_i = \frac{|\rho_{i,0} - \rho_{i,1}|}{\sqrt{\hat{p}(1 - \hat{p})(1/n_0 + 1/n_1)}}, \quad \hat{p} = \frac{c_{i,0} + c_{i,1}}{n_0 + n_1}. \quad (8)$$

A component is CONTROLLED only if its change rate depends significantly on the action; CLOCK/STATIC components change deterministically regardless of action; NOISE components change at a rate independent of the action, and are excluded from the value-estimation key below. This test lets VCBM recover the watch-repair environment’s causal structure (Section 5) without hand-specified labels.

### 4.1.3 Bookmark discovery and value estimation

A state is a *bookmark opportunity* if its projection onto CONTROLLED and CLOCK components matches a context previously observed immediately before a CONTROLLED component changed. At each bookmark, a key is formed from the STATIC, CONTROLLED, and CLOCK components only (marginalising out NOISE), and a Welford online estimator [37] accumulates, per (key, action), the mean and variance of the *full* episode return  $G(\tau)$ :

$$\mu_n = \mu_{n-1} + \frac{G(\tau) - \mu_{n-1}}{n}, \quad M_{2,n} = M_{2,n-1} + (G(\tau) - \mu_{n-1})(G(\tau) - \mu_n), \quad (9)$$

with standard error  $SE = \sqrt{M_{2,n}/(n(n-1))}$ . Marginalising out noise means samples from episodes that differ only in noise trajectory are correctly pooled, sample sharing unavailable to a plain per-state value table.

### 4.1.4 Action selection

At a bookmark, actions are sampled uniformly until both reach  $n_{\min}$  visits; thereafter a two-sample  $z$ -test on the Welford means,  $z = |\mu_0 - \mu_1|/\sqrt{SE_0^2 + SE_1^2}$ , determines significance. If significant, the higher-mean action is chosen greedily (residual exploration  $\epsilon_{\min}$ ); otherwise an  $\epsilon$ -greedy rule with decaying  $\epsilon$  continues exploring, in place of a goal-directed exploration curriculum [38], [39] unneeded in a two-action setting. At non-bookmark timesteps, an action is drawn uniformly, since Equation (7) guarantees such steps do not causally affect the outcome once the classifier has converged.

## 4.2 VCBM for language-based MDPs: retrieval-weighted credit blending

### 4.2.1 Motivation

The causal-role test of Section 4.1 requires a fixed, low-dimensional, fully observed state whose components can be tracked individually. An LLM agent (a Transformer-based [40] policy) operating in AppWorld [3] instead observes unstructured natural-language tool outputs from a partially observed environment, of the kind addressed more generally by deep variational RL for POMDPs [41], with no fixed component decomposition and no tractable per-component hypothesis test online. The second instantiation therefore replaces statistical causal *discovery* with a structural *heuristic* for what counts as a bookmark, and exact-match lookup with similarity-based *retrieval*, since no two AppWorld episodes share an identical state trajectory.

#### 4.2.2 Bookmark detection and episodic memory

A turn is bookmarked if it issues a state-mutating tool call, detected via the environment server’s execution interface with no extra network calls. Each bookmarked turn’s tool-output text  $x$  is embedded with a deterministic, dependency-free hashing-trick bag-of-words encoding  $g_\phi$  into a fixed-dimension vector,

$$g_\phi(x)_j = \frac{1}{\|v\|_2} \sum_{\text{token} \in x} \mathbb{1}[h(\text{token}) = j], \quad h(\text{token}) = \text{int}(\text{MD5}(\text{token})[:8]) \bmod d, \quad (10)$$

and stored in episodic memory as  $(\text{turn\_idx}, g_\phi(x), G(\tau))$ : position, embedding, and trajectory return.

#### 4.2.3 Retrieval and temperature-scaled weighting

Given a trajectory’s terminal-observation embedding  $z_H = g_\phi(x_H)$ , credit is distributed across the top- $k$  most similar bookmarks by cosine similarity: dense retrieval in the family of dense passage retrieval for text [42], but over bookmarked trajectory turns rather than a text corpus,

$$\text{sim}(z_H, k_i) = \frac{z_H \cdot k_i}{\|z_H\| \|k_i\|}, \quad w_i = \frac{\exp(\text{sim}_i/\tau)}{\sum_{j \in \text{top-}k} \exp(\text{sim}_j/\tau)}, \quad (11)$$

with temperature  $\tau$  controlling how sharply weight concentrates on the most similar bookmark.

Tokens in a retrieved bookmarked turn receive weight  $w_i$ ; all other tokens receive a fixed floor weight  $\alpha \in (0, 1)$ , down-weighted but never excluded; the weight vector is renormalised to sum to the output-token count, preserving overall gradient scale.

#### 4.2.4 Blending with the trajectory-level LOO advantage

Rather than replacing LOOP’s leave-one-out advantage outright, the per-token VCBM weight  $\tilde{w}_t$  (Equation (11), renormalised) is combined with the uniform LOO advantage  $A^{(k)}$  via a mixing coefficient  $\beta \in [0, 1]$ :

$$A_{\text{VCBM},t} = A^{(k)} \cdot \tilde{w}_t, \quad A_{\text{blend},t} = (1 - \beta) A^{(k)} + \beta A_{\text{VCBM},t}, \quad (12)$$

which reduces exactly to plain LOOP at  $\beta = 0$  and to a fully bookmark-concentrated advantage at  $\beta = 1$ ;  $A_{\text{blend},t}$  substitutes directly for the uniform  $A_t$  in Equation (6). Rollouts with zero bookmarked turns receive the unmodified scalar LOO advantage, since Equation (12) is undefined without a retrieval candidate.

### 4.3 Where the two instantiations agree and where they necessarily diverge

Both instantiations follow the detect/store/weight template and share the hypothesis that concentrating credit on causally relevant decision points outperforms an undifferentiated trajectory-level

signal, but they are not the same algorithm. Table 1 makes the divergence explicit: the tabular instantiation *discovers* bookmarks via a statistical test, with no learned embedding or similarity search; the AppWorld instantiation *assumes* a structural rule and relies on embedding-space retrieval to generalise across episodes that never share an identical state. The two are two different answers to the same design question (how to concentrate credit when discovery is or is not statistically tractable), evaluated independently in each regime, rather than one method validated twice.

**Table 1.** Mechanistic comparison of the two VCBM instantiations studied in this dissertation.

| Aspect                       | Tabular (Section 4.1)                       | AppWorld/LLM (Section 4.2)                 |
|------------------------------|---------------------------------------------|--------------------------------------------|
| Bookmark detection           | Statistical causal-role test<br>(Eq. (7))   | Structural rule (state-mutating tool call) |
| Memory content               | Welford $(\mu, \sigma^2)$ per (key, action) | Raw (index, embedding, return) tuples      |
| Retrieval mechanism          | Exact key match                             | Cosine-similarity top- $k$ (Eq. (11))      |
| Underlying policy            | Tabular $Q$ -style controller               | LoRA-adapted LLM, PPO-clipped<br>(Eq. (6)) |
| Fallback when data is scarce | Balanced random sampling                    | Flat floor weight $\alpha$                 |

#### 4.4 Six structural capabilities VCBM has that a uniform trajectory-level baseline lacks

The detect/store/weight template gives both instantiations six capabilities LOOP’s leave-one-out advantage (Equation (5)) structurally lacks, being a single scalar  $A^{(k)}$  applied uniformly with no notion of *which* token mattered:

- *Fine-grained credit*: the causal-role test (Eq. (7)) and AppWorld bookmark rule identify specific credit-bearing timesteps, expected to lower gradient variance and reduce reinforcement of spurious correlations.
- *Per-decision value estimates*: Welford (Eq. (9)) and retrieval-softmax (Eq. (11)) key weight to an individual (bookmark, action) or turn, not the whole rollout, useful where one decision is pivotal, as in watch-repair’s  $t=0$  choice.
- *Causal/structural knowledge*: only the tabular case has this genuinely (Eq. (7) separates CONTROLLED from independently-changing components); AppWorld substitutes a weaker, assumed heuristic. Uniquely verifiable against ground truth (Section 5.1).
- *Persistent memory*: the Welford accumulator and AppWorld memory bank carry information across iterations, unlike LOOP’s baseline, recomputed and discarded each batch.
- *Retrieval-based generalisation* (AppWorld only): Eq. (11) matches a trajectory to similar past bookmarks by cosine similarity rather than only the  $K - 1$  others in its batch, most useful where exact

scenario repeats are rare.

- *Selective, weighted credit*: the tabular key marginalises out NOISE; the AppWorld blend concentrates weight on bookmarks while down-weighting (not eliminating) the rest via floor  $\alpha$ , deliberately non-uniform, unlike LOOP’s identical  $A^{(k)}$  per token.

These are design rationale, not uniformly confirmed results. Only the first three are empirically observed, and only in the tabular setting: VCBM’s lower-mean, lower-variance episodes-to-converge (Table 2) and exact causal-structure recovery are confirmed significant (Mann–Whitney  $U = 19$ ,  $p < 0.001$ ,  $r = 0.91$ ). The remaining three are structurally present in AppWorld by construction and directionally consistent with the single-seed comparison (Section 5.2), but not yet statistically confirmed there; a second-seed AppWorld run (Section 6) would be the natural next step to test them.

## 5 Experiments

### 5.1 Controlled evaluation: RUDDER vs. VCBM in a delayed-reward tabular MDP

#### 5.1.1 Environment

The watch-repair environment models a single consequential decision under delayed, stochastic feedback. At  $t = 0$  the agent observes a random watch BRAND  $b \sim \text{Uniform}\{0, 1, 2, 3\}$  and chooses to repair it ( $a = 0$ , stochastic cost  $r^{\text{repair}} \sim \mathcal{N}(\mu_{\text{repair}}[b], \sigma_{\text{repair}}^2[b])$ ) or not ( $a = 1$ , no cost). For the remaining 50 timesteps, two independent NOISE counters increment at fixed, action-independent rates and a deterministic CLOCK advances; the action has no further effect after  $t = 0$ . Only a repaired watch yields a terminal reward at  $t = 50$ , the brand’s resale price minus accumulated transport costs and a stochastic penalty. Prices, repair costs, and transport-cost sensitivities differ by brand, so the optimal policy is brand-dependent (repairing is optimal only for brands 1 and 2), a known ground truth used purely for evaluation, never training.

#### 5.1.2 Method and metrics

Both RUDDER (Section 2.2, redistributing reward into a tabular direct- $Q$  update) and VCBM (Section 4.1) were trained for 20 independent seeds, terminated once the moving average (750-episode window) fraction of optimal first-decisions reached 95%. Table 2 reports episodes-to-convergence, our primary sample-efficiency metric [43], alongside converged decision quality.

**Table 2.** Watch-repair environment: episodes required to reach 95% correct first-decisions, 20 seeds each. RUDDER statistics computed from the 20 individual per-seed training logs (Section 5.1); VCBM statistics taken from the method’s own multi-seed summary log.

| Method | Seeds converged | Episodes to<br>converge (range) ↓ | Episodes to<br>converge (mean ±<br>std) ↓ | Final<br>good-decision % ↑                      |
|--------|-----------------|-----------------------------------|-------------------------------------------|-------------------------------------------------|
| RUDDER | 20/20           | 1,603 – 10,936                    | 3,327.1 ± 2,010.9<br>(median 2,829)       | 95.07% (by<br>construction of<br>stopping rule) |
| VCBM   | 20/20           | 1,196 – 7,757                     | 1,532.3 ± 1,428.0<br>(median 1,204.5)     | 95.07 ± 0.00%                                   |

### 5.1.3 Discovered causal structure

Across all 20 VCBM seeds, the causal-role classifier (Equation (7)) recovered an identical label assignment for the five state components (repaired-flag, two transport counters, brand, timestep): CONTROLLED, NOISE, NOISE, STATIC, CLOCK. This exactly matches the environment’s true generative structure: only the repaired-flag’s change rate depends on the action, the transport counters change at fixed rates, the brand is fixed, and the timestep is deterministic. This is direct evidence that VCBM’s discovery mechanism recovers ground-truth causal structure from interaction data alone, without being told which of the five components matters.

### 5.1.4 Interpretation

Both methods reach 95% in all 20 seeds, but VCBM’s episodes-to-converge distribution has a lower minimum (1,196 vs. 1,603) and maximum (7,757 vs. 10,936) than RUDDER’s, consistent with concentrating the signal on the causally relevant decision point converging at least as fast, with less tail risk, than an LSTM-based redistribution over the full trajectory. Because both distributions are right-skewed and non-normal, an unpaired two-sided Mann–Whitney  $U$  test (normal approximation, tie correction) rejects the null that the two distributions are identical ( $U = 19$ ,  $z = -4.90$ ,  $p < 0.001$ ), with a large effect size ( $r = 0.91$ ), confirming VCBM’s lower episodes-to-converge is statistically significant rather than sample noise.

## 5.2 Large-scale evaluation: LOOP vs. VCBM-blended LOOP on AppWorld

### 5.2.1 Setup

Both runs share Qwen2.5-7B-Instruct as the base policy, LoRA [44] rank 8 / alpha 16 on all seven attention and MLP projections, bf16 precision, FSDP2 training across 2 GPUs (a single-node instance of the distributed actor-learner pattern used at larger scale by IMPALA [45]), AdamW [46] at a constant learning rate of  $5 \times 10^{-5}$ , 200 iterations, the AppWorld train split, sparse per-episode reward, and up to 50 interactions per episode. The only intended difference is the credit mechanism: the baseline (csf3\_7b\_2gpu\_90iter) uses plain LOOP ( $\beta = 0$ ), and the comparison (csf3\_7b\_2gpu\_90iter\_vcbm\_r2) uses VCBM-blended LOOP ( $\beta = 1$ ,  $\tau = 0.5$ ,  $\text{top-}k = 6$ ,  $\alpha = 0.05$ ).

**Provenance and confidence caveat.** Final-logged-step figures below are exact, from the W&B summary export for each run (wandb Config.csv). Run-peak figures (Table 4) and outlier-gradient timing are chart-read from each run’s full per-iteration W&B dashboard export (ML-LOOP\_Graphs.pdf, VCBM\_Graphs.pdf, 223×213-pixel rendered tiles, no numeric time-series export available), so they carry an estimated  $\pm 10$ –20% reading uncertainty and are marked ~. Exact claims are single-seed, final-step results; chart-read claims are directionally informative but not precise; neither is statistically tested in the sense of Section 5.1’s Mann–Whitney test.

### 5.2.2 Task performance

Table 3 reports the headline task-completion metrics: mean episode return and success rate, both overall and split by AppWorld’s three difficulty tiers.

**Table 3.** AppWorld task performance, final logged training step. All figures are exact, taken directly from the W&B summary export for each run.

| Metric                                | LOOP   | VCBM   | Change    |
|---------------------------------------|--------|--------|-----------|
| Mean return (avg_return) ↑            | 0.6682 | 0.7300 | +9.3%     |
| Success rate, all difficulties ↑      | 0.3542 | 0.4583 | +29.4%    |
| Success rate, difficulty-averaged ↑   | 0.2569 | 0.3333 | +29.7%    |
| Success rate, difficulty 1 (easy) ↑   | 0.5833 | 0.5000 | −14.3%    |
| Success rate, difficulty 2 (medium) ↑ | 0.1875 | 0.5000 | +166.7%   |
| Success rate, difficulty 3 (hard) ↑   | 0.0000 | 0.0000 | no change |

Both runs solve zero difficulty-3 tasks ( $\approx 17\%$  of rollouts). The gain is concentrated almost entirely in difficulty 2 (+166.7%); difficulty 1 is unexpectedly slightly *lower* for VCBM (−14.3%), the only metric favouring LOOP.

### 5.2.3 Off-policy drift and optimisation stability

Because `epochs_per_iteration=2` with `strictly_on_policy=false`, each rollout batch is used for two gradient passes, so the second is off-policy w.r.t. the rollout-generating weights; the importance weight  $w_t$  (Section 2.3) quantifies the resulting policy drift. Table 4 reports the maximum importance weight and KL estimate at the final logged step (exact) alongside their run-peak value (chart-read; see the provenance caveat above).

**Table 4.** AppWorld off-policy drift and stability indicators. Final-step figures are exact (W&B summary export); run-peak figures (~) are chart-read from the per-run W&B dashboard export and carry an estimated  $\pm 10\text{--}20\%$  reading uncertainty.

| Metric                             | LOOP  | VCBM  | Change |
|------------------------------------|-------|-------|--------|
| Max importance weight, final ↓     | 3.97  | 3.09  | −22.2% |
| Max importance weight, peak ↓      | ~49   | ~14   | ~−71%  |
| Max log-prob. diff., peak ↓        | ~12.5 | ~5.5  | ~−56%  |
| KL estimate, final ↓               | 0.561 | 0.583 | +4.1%  |
| KL estimate, peak (abort =20) ↓    | ~18.5 | ~18.5 | ~tied  |
| Cumulative outlier-gradient events | 2     | 2     | tied   |

Mean probability drift is essentially identical between runs; the maximum importance weight is lower for VCBM at both the final step (−22.2%) and run peak (~14 vs. ~49), consistent with concentrating gradient mass on fewer tokens reducing worst-case drift. The KL estimate, however, is marginally higher for VCBM at the final step (+4.1%) and essentially tied at the run peak (both ~18.5, against the 20-nats abort threshold): a mixed rather than uniformly favourable result. Both runs recorded exactly two cumulative outlier-gradient events [47], [48]; LOOP’s second occurs around iteration ~95–100, VCBM’s later, ~150–160 (chart-read).

### 5.2.4 Efficiency

Table 5 reports output-token usage and a derived reward-per-token efficiency figure, since shorter trajectories are only a genuine efficiency gain if success rate does not fall.

**Table 5.** AppWorld generation efficiency, final logged training step. All figures are exact, taken directly from the W&B summary export for each run.

| Metric                           | LOOP    | VCBM    | Change |
|----------------------------------|---------|---------|--------|
| Mean output tokens per rollout ↓ | 2,001.6 | 1,625.2 | −18.8% |
| Return per 1,000 output tokens ↑ | 0.334   | 0.449   | +34.6% |

VCBM’s token count is 18.8% lower than LOOP’s while its success rate is simultaneously higher (Table 3), so the reduction reflects solving tasks in fewer steps rather than terminating early. *Return per*

1,000 output tokens, computed as  $\text{avg\_return}/(\text{avg\_output\_tokens}/1000)$ , is 34.6% higher for VCBM, the largest relative improvement in this section.

### 5.2.5 Parity and open questions

Both runs completed 200 iterations, 9,600 rollouts, the same base model, and the same learning rate. Diffing the exported W&B configurations (213 fields) shows they are *not* a clean single-field ablation: three VCBM-specific hyperparameters differ simultaneously ( $\tau$ : 1 vs. 0.5; top- $k$ : 4 vs. 6;  $\alpha$ : 0.1 vs. 0.05), alongside  $\beta = 1$  for VCBM. Since LOOP does not consume these fields, this is not a confound, but a comparison against one specific choice of hyperparameters, not a sweep. Two discrepancies remain unresolved in LOOP’s own configuration: its logged rollout count (48) does not follow arithmetically from its stated settings, and `llm.temperature=0` would imply zero advantage everywhere, contradicting the observed non-zero `adv_filtered_fraction`.

### 5.2.6 Interpretation

VCBM shows a higher mean return, higher success rate, lower maximum importance weight, and better token efficiency than LOOP; only the KL estimate does not favour it. As this is a single-seed, partly chart-read comparison, none of these differences have been tested for significance the way the tabular result was.

## 6 Conclusions

This dissertation tested whether RL credit assignment can be improved by explicitly identifying causally relevant decision points, rather than relying solely on trajectory-level baselines (LOOP) or learned return redistribution (RUDDER), via two instantiations jointly termed VCBM. In the tabular watch-repair environment, VCBM recovered the true causal structure across all 20 seeds and converged with a narrower, lower episode-count distribution than RUDDER, confirmed statistically significant (Mann–Whitney  $U = 19$ ,  $p < 0.001$ ,  $r = 0.91$ ; Section 5.1), directly supporting the central hypothesis where ground truth is fully verifiable. In AppWorld, VCBM-blended LOOP showed a higher mean return, higher success rate, lower final-step maximum importance weight, and better token efficiency, against a marginally higher final-step KL estimate (Section 5.2); these figures are exact but single-seed, so no significance test has been or can yet be run, and the AppWorld result is reported at the confidence the evidence supports: directionally consistent with the central hypothesis, but not yet statistically confirmed.

Two limitations remain. The AppWorld bookmark detector is a hand-specified structural rule, not a learned criterion, so it may not generalise where causally relevant actions are not tool calls; a

learned detector trained on the tabular classifier’s data (Equation (7)) would close this gap (Table 1). Memory-R2’s LoGo-GRPO [30] also uses memory to shape credit for memory-augmented LLM agents, and while Section 3.3 argues a mechanistic distinction, this has not been verified empirically. Both, alongside a second-seed AppWorld comparison, are left for future work.

## References

- [1] J. A. Arjona-Medina et al., ‘RUDDER: Return decomposition for delayed rewards,’ *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, pp. 13 544–13 555, 2019, Title/authors/affiliation verified against source PDF ([31] in Reference Documents/, arXiv:1806.07857v3); page range verified via the NeurIPS 2019 proceedings page and cross-checked against DBLP. (cited on pp. III, 1, 2).
- [2] K. Chen et al., *Reinforcement learning for long-horizon interactive LLM agents*, Introduces LOOP (leave-one-out PPO variant for interactive digital agents); verified from source PDF ([3] in Reference Documents/). Distinct from the PPO paper (schulman2017ppo), which main.tex previously miscited for LOOP., 2025. arXiv: 2502.01600 (cited on pp. III, 1, 3).
- [3] H. Trivedi et al., ‘AppWorld: A controllable world of apps and people for benchmarking interactive coding agents,’ in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Author list order and page range verified via ACL Anthology (aclanthology.org/2024.acl-long.850/), which matches the order used here. ACL 2024 Best Resource Paper., 2024, pp. 16 022–16 076 (cited on pp. III, 1, 7).
- [4] V. Mnih et al., ‘Human-level control through deep reinforcement learning,’ *Nature*, vol. 518, no. 7540, pp. 529–533, 2015 (cited on p. 1).
- [5] T. P. Lillicrap et al., ‘Continuous control with deep reinforcement learning,’ in *International Conference on Learning Representations (ICLR)*, Source: [19] in Reference Documents/, arXiv:1509.02971. Cited in Section 1 as a continuous robotic control example of the credit-assignment problem., 2016 (cited on p. 1).
- [6] D. Silver et al., *Mastering chess and shogi by self-play with a general reinforcement learning algorithm*, Source: [26] in Reference Documents/. Cited in Section 1 as a self-play game-playing example of the credit-assignment problem., 2017. arXiv: 1712.01815 (cited on p. 1).
- [7] P. F. Christiano et al., ‘Deep reinforcement learning from human preferences,’ *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 4299–4307, 2017, Title/authors verified against source PDF ([4] in Reference Documents/, arXiv:1706.03741v4); page range verified via DBLP (dblp.org/rec/conf/nips/ChristianoLBMLA17.html). (cited on p. 1).
- [8] Y. Bai et al., *Training a helpful and harmless assistant with reinforcement learning from human feedback*, Confirmed correct: this arXiv:2204.05862 paper is Anthropic’s RLHF methodology paper, distinct from Constitutional AI: Harmlessness from AI Feedback (Bai et al. 2022b, arXiv:2212.08073), which is a separate, later RLAIIF-based paper – verified the two are not the same paper before resolving which was intended here., 2022. arXiv: 2204.05862 (cited on p. 1).

- [9] L. Ouyang et al., ‘Training language models to follow instructions with human feedback,’ *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 27 730–27 744, 2022, Page range verified via the NeurIPS 2022 proceedings page. (cited on p. 1).
- [10] D. Amodei et al., *Concrete problems in AI safety*, Source: [1] in Reference Documents/. Cited in Section 1 for the reward-hacking/spurious-correlation framing., 2016. arXiv: 1606.06565 (cited on p. 1).
- [11] S. Chang et al., *Invariant rationalization*, Source: [2] in Reference Documents/. Cited in Section 1 for the reward-hacking/spurious-correlation framing., 2020. arXiv: 2003.09772 (cited on p. 1).
- [12] R. S. Sutton and A. G. Barto, ‘Reinforcement learning: An introduction,’ 2018 (cited on p. 1).
- [13] R. S. Sutton et al., ‘Policy gradient methods for reinforcement learning with function approximation,’ *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 12, pp. 1057–1063, 1999, Page range verified via the NeurIPS 1999 proceedings page and DBLP. (cited on p. 2).
- [14] A. Y. Ng, D. Harada and S. Russell, ‘Policy invariance under reward transformations: Theory and application to reward shaping,’ *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 278–287, 1999, Page range verified via DBLP ([dblp.org/rec/conf/icml/NgHR99.html](http://dblp.org/rec/conf/icml/NgHR99.html)); *Proceedings of the Sixteenth International Conference on Machine Learning*, Morgan Kaufmann. (cited on p. 2).
- [15] J. Schulman et al., *High-dimensional continuous control using generalized advantage estimation*, Confirmed as the ICLR 2016 conference paper (published as a conference paper at ICLR 2016, per the paper’s own header); ICLR has no separate proceedings page range distinct from its arXiv/OpenReview hosting, so the arXiv identifier remains the correct citable source., 2016. arXiv: 1506.02438 (cited on p. 2).
- [16] M. Minsky, ‘Steps toward artificial intelligence,’ *Proceedings of the IRE*, vol. 49, no. 1, pp. 8–30, 1961, Earliest widely-cited statement of the temporal credit-assignment problem in RL/AI. (cited on p. 2).
- [17] S. Hochreiter and J. Schmidhuber, ‘Long short-term memory,’ *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997 (cited on p. 2).
- [18] V. Mnih et al., ‘Asynchronous methods for deep reinforcement learning,’ in *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, ser. JMLR: W&CP, Source: [21] in Reference Documents/, arXiv:1602.01783. Page range verified via PMLR volume 48. Cited in Section 2.3 as actor-critic precedent for LOOP’s value-function-critic discussion., vol. 48, New York, NY, USA, 2016, pp. 1928–1937 (cited on p. 3).

- [19] J. Schulman et al., ‘Trust region policy optimization,’ in *Proceedings of the 31st International Conference on Machine Learning (ICML)*, ser. JMLR: W&CP, Source: [24] in Reference Documents/, arXiv:1502.05477. Page range verified via PMLR volume 37. Cited in Section 2.3 as trust-region-method precedent for LOOP’s value-function-critic discussion., vol. 37, Lille, France, 2015, pp. 1889–1897 (cited on p. 3).
- [20] W. Kool, H. van Hoof and M. Welling, ‘Buy 4 REINFORCE samples, get a baseline for free!’ *ICLR 2019 Workshop on Deep Reinforcement Learning Meets Structured Prediction*, 2019, Closely related leave-one-out baseline result underlying LOOP’s advantage estimator (Section 2.3). Exact workshop name verified (New Orleans, LA, 6 May 2019) against the author’s own poster page (wouterkool.github.io/pdf/poster-buy-4-samples-free-baseline.pdf). (cited on p. 3).
- [21] J. Schulman et al., *Proximal policy optimization algorithms*, Fields fully verified against source PDF ([25] in Reference Documents/, arXiv:1707.06347v2); arXiv-only, no separate venue printed, confirming the @Misc entry type is correct., 2017. arXiv: 1707.06347 (cited on p. 3).
- [22] A. Harutyunyan et al., *Hindsight credit assignment*, Verified via arXiv abstract page (submitted 5 Dec 2019); Advances in Neural Information Processing Systems 33 (NeurIPS 2019)., 2019. arXiv: 1912.02503 (cited on p. 3).
- [23] T. Mesnard et al., *Counterfactual credit assignment in model-free reinforcement learning*, Verified via arXiv abstract page (submitted 18 Nov 2020, revised 14 Dec 2021); reported elsewhere as ICML 2021, but the exact proceedings venue is not confirmed from the arXiv abstract page itself., 2020. arXiv: 2011.09464 (cited on p. 4).
- [24] A. Meulemans et al., *Would I have gotten that reward? Long-term credit assignment by counterfactual contribution analysis*, Verified via arXiv abstract page (submitted 29 Jun 2023, revised 31 Oct 2023); NeurIPS 2023 spotlight., 2023. arXiv: 2306.16803 (cited on p. 4).
- [25] Z. Ren et al., *Learning long-term reward redistribution via randomized return decomposition*, Verified via arXiv abstract page (submitted 26 Nov 2021, revised 17 Mar 2022); ICLR 2022 spotlight., 2021. arXiv: 2111.13485 (cited on p. 4).
- [26] M. Li, K.-Z. Lee and E. Bareinboim, *Counterfactual shapley credit assignment*, Verified via arXiv abstract page (submitted 18 Jul 2026, revised 7 Aug 2026); Reinforcement Learning Journal 2026 (also RLC 2026). Referred to informally as “ $\phi$ -PPO” in main.tex., 2026. arXiv: 2607.16999 (cited on p. 4).
- [27] Q. Wei et al., *Reinforcing multi-turn reasoning in LLM agents via fine-grained reward structure and credit assignment*, Verified via arXiv abstract page (submitted 17 May 2025, revised 21 Aug 2026). Verified title differs from the informal “Multi-turn GRPO” shorthand used for this key., 2025. arXiv: 2505.11821 (cited on p. 4).

- [28] Y. Wang et al., *Group-graph policy optimization for long-horizon agentic reinforcement learning*, Verified via arXiv abstract page (submitted 22 Jun 2026); builds a global state-transition graph and evaluates on WebShop, ALFWorld, and AppWorld., 2026. arXiv: 2606.22995 (cited on p. 4).
- [29] H. Wang et al., *SPA-RL: Reinforcing LLM agents via stepwise progress attribution*, Verified via arXiv abstract page (submitted 27 May 2025)., 2025. arXiv: 2505.20732 (cited on p. 4).
- [30] S. Yan et al., *Memory-R2: Fair credit assignment for long-horizon memory-augmented LLM agents*, Verified via arXiv abstract page (submitted 20 May 2026); introduces LoGo-GRPO. LoGo-GRPO’s memory-augmented credit mechanism is architecturally close to VCBM’s episodic-memory blend (Section 4.2); the Related Work section discusses this proximity honestly rather than overclaiming distinctiveness., 2026. arXiv: 2605.21768 (cited on pp. 4, 15).
- [31] E. Pignatelli et al., ‘A survey of temporal credit assignment in deep reinforcement learning,’ *Transactions on Machine Learning Research*, 2024, Verified via arXiv abstract page (arXiv:2312.01072, submitted 2 Dec 2023, revised 4 Jul 2024) and independent confirmation of TMLR 2024 acceptance. (cited on p. 5).
- [32] G. Zhang et al., *The landscape of agentic reinforcement learning for LLMs: A survey*, Verified via arXiv abstract page (submitted 2 Sep 2025); accepted to Transactions on Machine Learning Research (TMLR), per the OpenReview forum listed (forum ID RY19y2RI1O)., 2025. arXiv: 2509.02547 (cited on p. 5).
- [33] C. Zhang, *From reasoning to agentic: Credit assignment in reinforcement learning for large language models*, Verified via arXiv abstract page (submitted 10 Apr 2026, revised 9 Aug 2026); organises 47 credit-assignment methods for LLM RL by granularity and methodology., 2026. arXiv: 2604.09459 (cited on p. 5).
- [34] M. Chen, L. Wang and B. Qu, *The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon LLM agents*, Verified via arXiv abstract page (submitted 7 Aug 2026); surveys long-horizon LLM-agent research including credit assignment across six lifecycle stages., 2026. arXiv: 2608.06663 (cited on p. 5).
- [35] T. Kipf, E. van der Pol and M. Welling, ‘Contrastive learning of structured world models,’ in *International Conference on Learning Representations (ICLR)*, Source: [15] in Reference Documents/, arXiv:1911.12247. Cited in Section 4.1 as structured world-model context., 2020 (cited on p. 6).
- [36] D. Ha and J. Schmidhuber, ‘Recurrent world models facilitate policy evolution,’ in *Advances in Neural Information Processing Systems (NeurIPS)*, Source: [6] in Reference Documents/, arXiv:1809.01999. 32nd Conference on Neural Information Processing Systems (NeurIPS 2018), Montréal, Canada. Page range verified via the NeurIPS 2018 proceedings page. Cited in Section 4.1

as world-model-learning context for structured state representation., vol. 31, 2018, pp. 2455–2467 (cited on p. 6).

- [37] B. P. Welford, ‘Note on a method for calculating corrected sums of squares and products,’ *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962, Source of the online mean/variance estimator used in Equation (9). (cited on p. 7).
- [38] C. Florensa et al., ‘Automatic goal generation for reinforcement learning agents,’ in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, Source: [9] in Reference Documents/, arXiv:1705.06366. Page range verified via PMLR volume 80. Cited in Section 4.1 for goal-directed exploration context., vol. 80, Stockholm, Sweden: PMLR, 2018, pp. 1515–1528 (cited on p. 7).
- [39] R. Mendonca et al., ‘Discovering and achieving goals via world models,’ in *Advances in Neural Information Processing Systems (NeurIPS)*, Source: [23] in Reference Documents/, arXiv:2110.09514. 35th Conference on Neural Information Processing Systems (NeurIPS 2021), Sydney, Australia. Page range verified via the ACM Digital Library record (DOI 10.5555/3540261.3542127). Cited in Section 4.1 as an example of goal-directed exploration for VCBM’s action-selection discussion., vol. 34, 2021, pp. 24 379–24 391 (cited on p. 7).
- [40] A. Vaswani et al., ‘Attention is all you need,’ in *Advances in Neural Information Processing Systems (NeurIPS)*, Source: [28] in Reference Documents/, arXiv:1706.03762. 31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA. Page range verified via the NeurIPS 2017 proceedings page. Cited in Section 4.2 for the LLM policy’s Transformer architecture., vol. 30, 2017, pp. 5998–6008 (cited on p. 7).
- [41] M. Igl et al., ‘Deep variational reinforcement learning for POMDPs,’ in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, Source: [11] in Reference Documents/, arXiv:1806.02426. Page range verified via PMLR volume 80. Cited in Section 4.2 for the partially-observed-setting context., vol. 80, Stockholm, Sweden: PMLR, 2018, pp. 2117–2126 (cited on p. 7).
- [42] V. Karpukhin et al., *Dense passage retrieval for open-domain question answering*, Source: [13] in Reference Documents/. Cited in Section 4.2 for VCBM’s cosine-similarity retrieval mechanism., 2020. arXiv: 2004.04906 (cited on p. 8).
- [43] S. Levine et al., *Offline reinforcement learning: Tutorial, review, and perspectives on open problems*, Source: [17] in Reference Documents/. Cited in Section 5.1 for the sample-efficiency metric framing., 2020. arXiv: 2005.01643 (cited on p. 10).
- [44] E. J. Hu et al., ‘LoRA: Low-rank adaptation of large language models,’ *International Conference on Learning Representations (ICLR)*, 2022, Underlies the LoRA-adapted policy used in the AppWorld/LOOP training pipeline (Section 5.2). (cited on p. 12).

- [45] L. Espeholt et al., *IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures*, Source: [5] in Reference Documents/. Cited in Section 5.2 as the distributed actor-learner training pattern context for FSDP2 training., 2018. arXiv: 1802.01561 (cited on p. 12).
- [46] D. P. Kingma and J. L. Ba, ‘Adam: A method for stochastic optimization,’ in *International Conference on Learning Representations (ICLR)*, Source: [14] in Reference Documents/, arXiv:1412.6980. Cited in Section 5.2 for the AdamW optimiser used in AppWorld/LOOP/VCBM training., 2015 (cited on p. 12).
- [47] S. S. Du et al., ‘Gradient descent can take exponential time to escape saddle points,’ in *Advances in Neural Information Processing Systems (NeurIPS)*, Source: [32] in Reference Documents/, arXiv:1705.10412. 31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA. Page range verified via the NIPS 2017 proceedings page. Cited in Section 5.2 for the outlier-gradient discussion., vol. 30, 2017, pp. 1068–1078 (cited on p. 13).
- [48] M. Hardt, B. Recht and Y. Singer, *Train faster, generalize better: Stability of stochastic gradient descent*, Source: [7] in Reference Documents/. Cited in Section 5.2 for the outlier-gradient/generalisation discussion., 2016. arXiv: 1509.01240 (cited on p. 13).

# Visual Causal Chain Bookmarking: Causal Credit Selection for Long-Horizon Agentic Reinforcement Learning

Prathap Selvakumar

Department of Electrical and Electronic Engineering, University of Manchester  
prathap.selvakumar@postgrad.manchester.ac.uk

June 2026

## Abstract

Long-horizon RL with sparse terminal rewards must distribute a single scalar reward across  $H$  actions, most of which have no causal bearing on the outcome. Existing credit assignment methods (RUDDER, GAE, LOOP, GRPO) redistribute credit temporally or uniformly but never ask *which* timesteps actually caused the outcome. We introduce **Visual Causal Chain Bookmarking (VCBM)**, which detects timesteps where an agent’s action produced a persistent environment-state change, retrieves the most relevant such events via embedding similarity to the terminal state, and restricts policy gradient updates to this subset. We prove the resulting estimator is unbiased and strictly reduces gradient variance relative to uniform credit assignment. We further introduce an HTTP request safety tracker giving a deterministic bound on rollout collection time, eliminating a measured  $\sim 14\%$  per-iteration baseline failure rate, and report wall-clock evidence that a naive causal-detector implementation incurs a 1596% rollout-collection overhead — almost entirely (98.0%) attributable to state-snapshot diffing — for which we specify a zero-extra-call, provably equivalent fix.

## 1. Problem

We extend the standard MDP to an **Agentic Interactive MDP**  $\mathcal{M}_{\mathcal{E}} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, R, \gamma, H, \mathcal{E})$ , where  $\mathcal{A}$  is natural language containing executable code and  $\mathcal{E}$  has execution time  $\Omega(a, \mathcal{E})$  with  $\sup \Omega = \infty$  (non-terminating code is possible). Reward is sparse terminal:  $R(s_t, a_t, s_{t+1}) = r_T \in [0, 1]$  at  $t=H$ , else 0.

**Def. (Causally Inert Timestep).** For state-change function  $f: \mathcal{S} \rightarrow \mathbb{R}^d$  and threshold  $\delta$ ,  $t$  is inert if  $\|f(s_t) - f(s_{t-1})\|_2 \leq \delta$ . Active set  $\mathcal{C}(\tau)$ , inert set  $\mathcal{I}(\tau)$ , and  $\text{CCR}(\tau) = |\mathcal{C}(\tau)|/H$ .

**Theorem 1 (Gradient Waste).** For  $\hat{g} = \sum_t A_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$  with  $\mathbb{E}[A_t] = 0$  on  $\mathcal{I}(\tau)$ :  $\text{Var}[\hat{g}] \geq \text{Var}[\sum_{t \in \mathcal{C}(\tau)} A_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$ , equality iff  $A_t = 0$  a.s. on  $\mathcal{I}(\tau)$ . *Proof:* decompose  $\hat{g} = \hat{g}_{\mathcal{C}} + \hat{g}_{\mathcal{I}}$ ; zero-mean inert-step terms add strictly positive variance unless degenerate.  $\square$  **Corollary:** LOOP and GRPO incur avoidable variance whenever  $\text{CCR}(\tau) < 1$ .

## 2. VCBM Mechanism

Five components: state encoder  $g_{\phi}$ , causal detector  $\varphi_{\delta}(s_t, s_{t-1}) = \mathbf{1}[\|f(s_t) - f(s_{t-1})\|_2 > \delta]$ , episodic memory  $\{k_i = g_{\phi}(s_{t_i}), v_i = (a_{t_i}, G_{t_i})\}$  keyed on detected events, retrieval by cosine similarity  $\rho_i$  to terminal state  $z_H$  with top- $k$  set  $\mathcal{R}_k(\tau)$ , and softmax credit weights  $w_t \propto \exp(\rho_i / \tau_{\text{temp}})$  on  $\mathcal{R}_k(\tau)$ , zero elsewhere. The gradient is  $\nabla_{\theta} J_{\text{VCC}} = \mathbb{E}_{\tau} [\sum_{t \in \mathcal{R}_k(\tau)} w_t A_{\text{VCC}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$ , replacing LOOP’s uniformly-broadcast leave-one-out scalar  $A_{\text{LOOP}}(\tau_i) = R_i - \frac{1}{N-1} \sum_{j \neq i} R_j$  with a sparse, relevance-weighted assignment. Final advantage blends  $A_{\text{blend}} = (1-\alpha)A_{\text{LOOP}} + \alpha w_t A_{\text{VCC}}$ ,  $\alpha=0$  recovering standard LOOP exactly.

**Theorem 2 (Unbiasedness).** If the detector has zero false-negative rate,  $\mathbb{E}_{\tau}[\hat{g}_{\text{VCC}}] = \nabla_{\theta} J(\theta)$ , since  $w_t$  is a  $\theta$ -independent post-hoc function and zero-weighted terms contribute exactly zero by the policy gradient theorem.

**Theorem 3 (Variance Reduction).**  $\text{Var}[\hat{g}_{\text{VCC}}] \leq \text{Var}[\hat{g}_{\text{LOOP}}] - \sum_{t \in \mathcal{I}(\tau)} \text{Var}[A_t^{\text{LOOP}} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$ , immediate from Theorem 1 since VCC zeros all inert-step weights. **Corollary:**  $N_{\text{VCC}} \leq N_{\text{LOOP}} \cdot \text{CCR}(\tau)$  in the zero-error limit.

## 3. HTTP Safety Tracker & Cost of Detection

A tracker  $\mathcal{H}$  enforces per-request timeout  $T_{\text{max}}$ , returning (TIMEOUT,  $-\epsilon, s_t$ ) if  $\Omega(a_t, \mathcal{E}) > T_{\text{max}}$ . **Theorem 4:**  $\Omega_{\mathcal{H}}(\tau) \leq H(T_{\text{max}} + T_{\text{gen}}) + T_{\text{overhead}}$  by induction over turns, giving  $P_{\text{collapse}} = 0$  — confirmed across 200 iterations with zero crashes, against a measured 14% baseline collapse rate.

The naive detector implementation reads a full state snapshot before/after every turn (`get_full_state_snapshot()`), doubling HTTP calls per turn. The one observed collapse incident was a malformed retry loop:

```
while True: try: x.extend(api.top_tracks()) except Exception: if e.code==403: break
```

the `e.code` attribute does not exist on the raised exception, so the generic `except` swallows the `ReadTimeout` and retries identically forever; this blocked for 243 s against a 100 s per-call timeout, requiring a forced `SIGKILL` restart.  $\mathcal{H}$  must wrap the *individual* HTTP call, not the code block, to bound this case. Measured cost over a matched 50-step window (Qwen2.5-1.5B-Instruct, LoRA rank 8, single RTX 4080):

| Phase                                          | LOOP (s)    | VCBM-naive (s) | Rel. cost     |
|------------------------------------------------|-------------|----------------|---------------|
| Model setup / log-probs / optim. / grad. steps | 143         | 125            | −13% avg      |
| <b>Rollout collection</b>                      | <b>88.5</b> | <b>1500</b>    | <b>+1596%</b> |
| Compute gradients / adv. filtering             | 73.6        | 62.1           | −16% avg      |
| <b>Total</b>                                   | <b>305</b>  | <b>1687</b>    | <b>+453%</b>  |

**Prop. 1:** rollout collection accounts for 98.0% of the total cost differential, isolating state-diffing as the dominant overhead source. Since AppWorld already tracks each request’s HTTP method and status, a zero-extra-call detector is available:  $\varphi^{\text{http}}(t) = \mathbf{1}[\exists \text{req} : \text{method} \in \{\text{POST}, \text{PUT}, \text{DELETE}, \text{PATCH}\} \wedge \text{status} < 400]$ . **Prop. 2 (Equivalence):** under REST conformance,  $\varphi^{\text{http}}(t) = \varphi_{\delta}(s_t, s_{t-1})$  for all  $t$ , so this is a drop-in replacement with no change to Theorems 2–3; projected speedup  $\sim 16\text{--}17\times$  on the dominant cost term, not yet re-measured at full scale.

#### 4. Four-Factor Validation Protocol (Proposed)

Define episodes-to-threshold  $N_{\theta}(\text{method}) = \min\{N : \frac{1}{100} \sum_{i=N-99}^N \mathbf{1}[\text{success}_i] \geq \theta\}$ ,  $\theta=0.90$ . **(F1) Convergence speed:** VCBM vs. RUDDER, 5 seeds, MiniGrid-KeyCorridorS3, report  $N_{0.90}^{\text{RUDDER}}/N_{0.90}^{\text{VCBM}}$ . **(F2) Horizon scaling:** repeat F1 across S3–S6 ( $H \approx 30$  to 96) with  $\text{CCR}(\tau)$  decreasing in  $H$ ; define  $\text{Gap}(H) = N_{0.90}^{\text{RUDDER}}(H) - N_{0.90}^{\text{VCBM}}(H)$  and test whether  $\text{Gap}(H)$  increases significantly via linear regression. **(F3) Revisit robustness:** a gridworld requiring  $\ell \in \{1, 3, 5\}$  revisits before the productive transition; VCBM vs. plain leave-one-out,  $\text{LoopGap}(\ell) = N_{0.90}^{\text{LOOP}}(\ell) - N_{0.90}^{\text{VCBM}}(\ell)$ , plus a qualitative *Loop Disambiguation Rate* (fraction of successful episodes where the retrieved chain identifies the productive revisit). **(F4) Significance:**  $\geq 5$  seeds throughout; Welch’s  $t$ -test with Cohen’s  $d$  for all  $N_{0.90}$  comparisons; regression slopes for  $\text{Gap}(H)/\text{LoopGap}(\ell)$  tested against zero; Bonferroni correction within each factor; no single-seed result reported as a finding.

**Ablation design.** The *LOOP-matched* condition (identical context length, rollout count, GPU allocation, but  $\alpha=0$ ) isolates the causal mechanism from configuration effects; *random-anchors* tests whether specific causal identification matters versus mere sparsification.

| Condition                    | $\alpha$ | Detector           | HTTP Tracker |
|------------------------------|----------|--------------------|--------------|
| LOOP-original / LOOP-matched | 0.0      | —                  | ✗            |
| VCBM-no-tracker              | 0.1      | ✓                  | ✗            |
| VCBM-random-anchors          | 0.1      | random             | ✓            |
| VCBM-uniform-credit          | 0.1      | ✓ (uniform $w_t$ ) | ✓            |
| VCBM-full                    | 0.1      | ✓                  | ✓            |

#### 5. Empirical Evidence & Status

Two independent 200-iteration runs (Qwen2.5-1.5B-Instruct, LoRA rank 8, 9.23M/1.55B trainable params, AdamW  $5 \times 10^{-5}$ , PPO  $\epsilon=0.1$ ,  $\alpha=0.1$ ,  $\text{top-}k=5$ ,  $\tau_{\text{temp}}=0.1$ ) confirm Theorem 4 directly: zero crashes in either configuration (4096-token eager-mode, 4 rollouts/iter; and 8192-token CUDA-graph-enabled, 8 rollouts/iter), no high-KL policy collapse (all  $\text{kl\_estimate} \in [-0.86, 0.78]$ ), and advantage-filtering fractions of 0.75–0.88 indicating room for sample efficiency gains via Corollary 1. The compiled, larger-cache configuration reduced rollout cost from 36.9 to 14.15 s/iteration despite collecting  $2\times$  the rollouts, attributable to CUDA graphs and KV-cache sizing rather than the detector itself. These results substantiate the safety claim (Theorem 4) and the cost-attribution claim (Prop. 1); the variance-reduction claim (Theorem 3) awaits the LOOP-matched ablation and four-factor protocol above, both specified but not yet run, alongside re-measurement under the corrected detector of Section 3.

**References:** Arjona-Medina et al., *RUDDER*, NeurIPS 2019 · Schulman et al., *GAE*, ICLR 2016 · Schulman et al., *PPO*, arXiv:1707.06347 · Apple ML Research, *LOOP*, 2025 · DeepSeek-AI, *GRPO*, 2024 · Andrychowicz et al., *HER*, NeurIPS 2017 · Pritzel et al., *Neural Episodic Control*, ICML 2017 · Trivedi et al., *AppWorld*, ACL 2024 · Qwen Team, *Qwen2.5 Technical Report*, arXiv:2412.15115 · Hu et al., *LoRA*, arXiv:2106.09685 · Loshchilov & Hutter, *AdamW*, ICLR 2019 · Kwon et al., *vLLM/PagedAttention*, SOSP 2023 · Zhao et al., *PyTorch FSDP*, VLDB 2023.

## General Risk Assessment Form

|                                |                                      |                                     |                                    |                              |                         |
|--------------------------------|--------------------------------------|-------------------------------------|------------------------------------|------------------------------|-------------------------|
| <b>Date: (1)</b><br>28/06/2026 | <b>Assessed by: (2)</b><br>Dr. Pawel | <b>Checked / Validated* by: (3)</b> | <b>Location: (4)</b><br>Manchester | <b>Assessment ref no (5)</b> | <b>Review date: (6)</b> |
|--------------------------------|--------------------------------------|-------------------------------------|------------------------------------|------------------------------|-------------------------|

**Task / premises: (7)** Working from home

| <b>Activity (8)</b> | <b>Hazard (9)</b>                                                                                           | <b>Who might be harmed and how (10)</b>                                                                                              | <b>Existing measures to control risk (11)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Risk rating (12)</b> | <b>Result (13)</b> |
|---------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------|
| Working from home   | Lone working                                                                                                | Home working staff<br><br>Isolated,                                                                                                  | <ol style="list-style-type: none"> <li>1. Please refer to the University Lone Working <a href="#">policy</a> and <a href="#">guidance</a> for more information</li> <li>2. Please refer to the new University <a href="#">Working at Home guidance</a></li> <li>3. Please refer to the new University <a href="#">Wellbeing Support</a> website</li> <li>4. Staff are able to have regular direct contact with line manager and colleagues via phone, Teams, Zoom or email</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Low                     | A                  |
| Working from home   | Poor posture, repetitive movements, eye strain, from long periods looking at DSE (display screen equipment) | Staff, students, visitors<br><br>Back strain (due to poor posture).<br>Repetitive Strain Injury (RSI) to upper limbs.<br>Eye strain. | <ol style="list-style-type: none"> <li>1. Please refer to the DSE <a href="#">policy</a>, <a href="#">guidance</a> and <a href="#">poster</a> for more information on how to set up your workstation properly</li> <li>2. Complete <a href="#">DSE Self-Assessment</a> for your home working at least every 2 years but sooner if any changes or pain is experienced.</li> <li>3. Complete <a href="#">Homeworking self-assessment checklist</a></li> <li>4. Set up workstation to a comfortable position with good lighting and natural light where possible</li> <li>5. Take regular breaks away from the screen</li> <li>6. Regularly stretch your arms, back, neck, wrists and hands to avoid repetitive strain injuries. Refer to seated <a href="#">exercises</a></li> <li>7. Set up a desktop working space where possible and try to avoid working on a laptop without a docking station, separate keyboard or mouse</li> <li>8. Small equipment purchases of up to £50 to assist with working from home, contact local administration for more details.</li> <li>9. If experiencing ill-health issues contact your local DSE assessor or local safety advisor who will perform a full DSE assessment.</li> <li>10. Occupational health referral where issues cannot be resolved from full DSE assessment.</li> <li>11. DSE users should have regular eye tests, follow <a href="#">guidance</a></li> <li>12. FSE run monthly DSE awareness sessions on Teams</li> </ol> | Low                     | A                  |

| Activity (8)                 | Hazard (9)                                                    | Who might be harmed and how (10)                                                                                              | Existing measures to control risk (11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Risk rating (12) | Result (13) |
|------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|
| Working from home            | Stress / Wellbeing                                            | Home working staff<br><br>Psychosocial effects<br>Work / Life imbalance<br>Anxiety<br>Poor performance<br>Fatigue & Tiredness | <ol style="list-style-type: none"> <li>1. Please refer to <a href="#">Stress Prevention and Management toolkit</a> for policies and guidance</li> <li>2. Please refer to new University guidance for <a href="#">Managing teams working from home</a></li> <li>3. Please refer to <a href="#">Seven rules of home working</a> published by AMBS</li> <li>4. Please refer to <a href="#">Guidance for Managers</a> and <a href="#">Guidance for Staff</a></li> <li>5. Complete training <a href="#">Work Related Stress: Identification, Prevention &amp; Management (Online)</a></li> <li>6. The <a href="#">University Stress Assessment tool</a> can be used to highlight the main factors for an individual that are recognised as having the potential to lead to work-related stress</li> <li>7. Projects, work plans and objectives to be discussed and agreed with line manager regularly</li> <li>8. Refer to full <a href="#">FSE Stress Risk Assessment</a></li> <li>9. Regular contact meetings with manager and peers via Teams, Zoom, email and phone</li> <li>10. Define working hours, set a start &amp; close daily routine, and prioritise your tasks.</li> <li>11. Individual may self-refer to <a href="#">Occupational Health Service</a> or to the <a href="#">Counselling and Mental Health Service</a></li> <li>12. Manager / Employee consultation, wellbeing focused</li> </ol> | Low              | A           |
| Use of electrical appliances | Misuse of electrical appliance, faulted electrical appliance. | Home working staff<br><br>Electric shock, burns and fire                                                                      | <ol style="list-style-type: none"> <li>1. All office equipment used in accordance with the manufacturer's instructions</li> <li>2. Visual checks before use to make sure equipment, cables and free from defects</li> <li>3. University IT equipment brought home should already be PAT tested, small electrical items can be tested through the <a href="#">FSE I&amp;F team</a></li> <li>4. The domestic electrical supply and equipment owned by the employee is the responsibility of the employee to maintain</li> <li>5. Liquid spills cleaned up immediately</li> <li>6. Defective plugs, cables and equipment should be taken out of use</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Med              | A           |

| Activity (8)                  | Hazard (9)                    | Who might be harmed and how (10)                                         | Existing measures to control risk (11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Risk rating (12) | Result (13) |
|-------------------------------|-------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|
| Moving around the home office | Obstructions and trip hazards | Home working staff<br><br>Slips, trips and falls causing physical injury | 1. Floors and walkways kept clear of items, e.g. boxes, packaging, equipment etc<br>2. Furniture is arranged such that movement of people and equipment are not restricted<br>3. Make sure all areas have good level of lighting<br>4. Reasonable standards of housekeeping maintained<br>5. Trailing cables positioned neatly away from walkways<br>6. Cabinet drawers and doors kept closed when not in use                                                                                                               | Med              | A           |
| Working from home             | Fire                          | Staff Home Working<br><br>Risk of burns, smoke inhalation, asphyxiation  | 1. In the event of a fire evacuate out of the building and call the fire brigade on 999<br>2. All waste, including combustible waste, removed regularly<br>3. Heaters located away from combustible materials and switched off when not in use, don't leave heaters unattended<br>4. Avoid daisy chaining and do not overload extension leads<br>5. Test smoke alarm routinely and replace batteries every 6-12 months<br>6. Please refer to fire brigade <a href="#">Home Fire Safety</a> and Smoke <a href="#">Alarms</a> | Med              | A           |
| Working from home             | High risk activities          | Staff Home Working<br><br>Personal injuries / accidents                  | 1. Only office activities with IT equipment and associated peripherals are carried out without further specific risk assessment                                                                                                                                                                                                                                                                                                                                                                                             | Low              | A           |

| Activity (8)      | Hazard (9)           | Who might be harmed and how (10)                                          | Existing measures to control risk (11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Risk rating (12) | Result (13) |
|-------------------|----------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|
| Working from home | Manual handling      | Staff Home Working<br><br>Back pain bruises, sprains, strains, fractures. | <ol style="list-style-type: none"> <li>1. When ordering goods the intended recipient must first check the weight and dimension of the delivery. Please use page 7 of the <a href="#">HSE weight guide</a> to help determine if the item is too large, bulky or heavy to be received at home by the recipient. Item should only be delivered to a home address if it is below 25kg and can be comfortably carried by the recipient</li> <li>2. Staff are trained via SLD courses (<a href="#">TLC0510</a> or <a href="#">TLCA500</a> as appropriate), and familiar with correct handling technique and seek assistance when needed</li> <li>3. Use kinetic lifting techniques e.g. feet apart, load held close and in front of the body. If lifting off the floor, bend knees and keep the spine neutral.</li> <li>4. Ensure there is a firm grip on the item whilst moving</li> <li>5. Ensure trip hazards are removed on route from the front door to where the item is to be located.</li> <li>6. Do not store large, heavy, fragile or cumbersome items at height (e.g. on high shelves or on top of cabinets/bookcases etc.)</li> </ol> | Low              | A           |
| Working from home | Accident Incidents / | Staff Home working<br><br>Injuries from home working activities           | <ol style="list-style-type: none"> <li>1. If you suffer an accident / incident whilst working at home in relation to your workstation, please report the event to your line manager and the School Safety Advisor and complete an <a href="#">accident / incident form</a>. Ensure you have adequate first aid supplies to treat minor injuries. Call 999 in an emergency</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Low              | A           |

#### Action plan (14)

| Ref No | Further action required | Action whom | by | Action when | by | Done |
|--------|-------------------------|-------------|----|-------------|----|------|
|        |                         |             |    |             |    |      |
|        |                         |             |    |             |    |      |

|                                                                                                                            |                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>I have read and I understand the content of this office risk assessment and I will comply with this Risk Assessment</b> |                                                                                    |
| <b>Name (print)</b>                                                                                                        | <b>Signature</b>                                                                   |
| Prathap Selvakumar                                                                                                         |  |
|                                                                                                                            |                                                                                    |